

What is high level information fusion and why is it important ?

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Biography

Dr. Claire Laudy

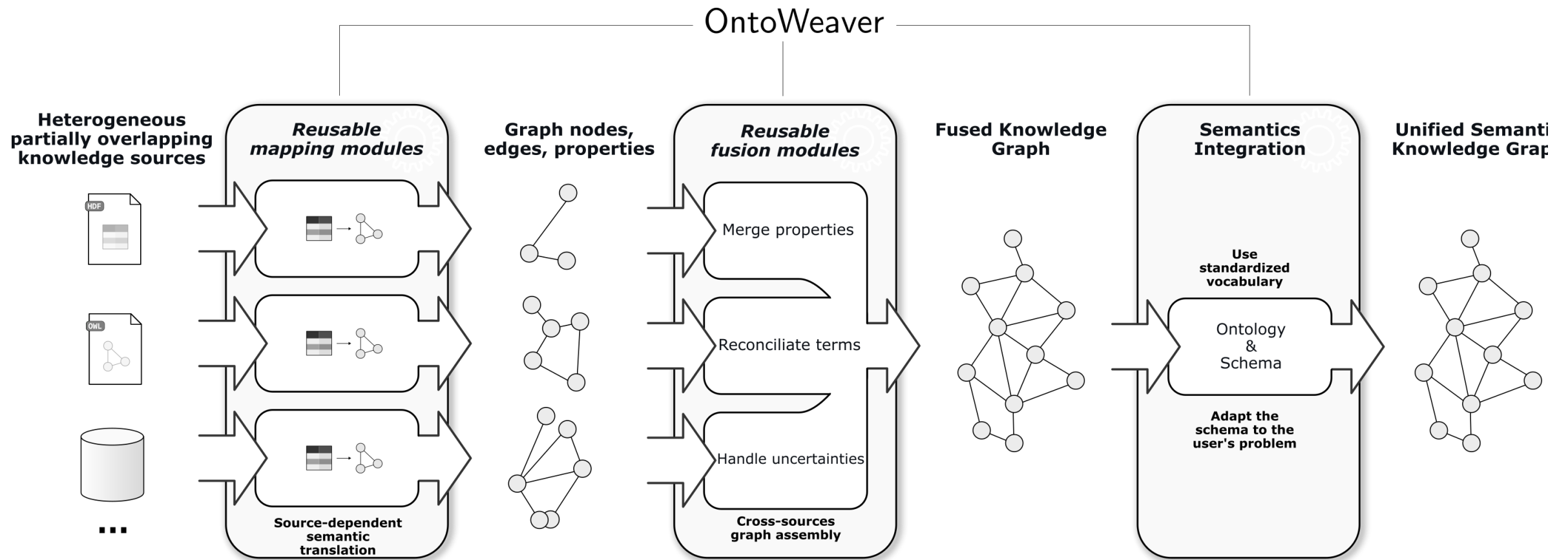
- Senior Research Engineer
- 25y experience
 - industry
- Symbolic & Explainable AI
 - Crisis management, Citizen Science, Cultural Heritage, Precision Medicine
- 2000-2025: CortAlx Labs, Thales
- 2025-2026: Institut Pasteur
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Need for Information Fusion in OntoWeaver

What/Why/How information fusion ?

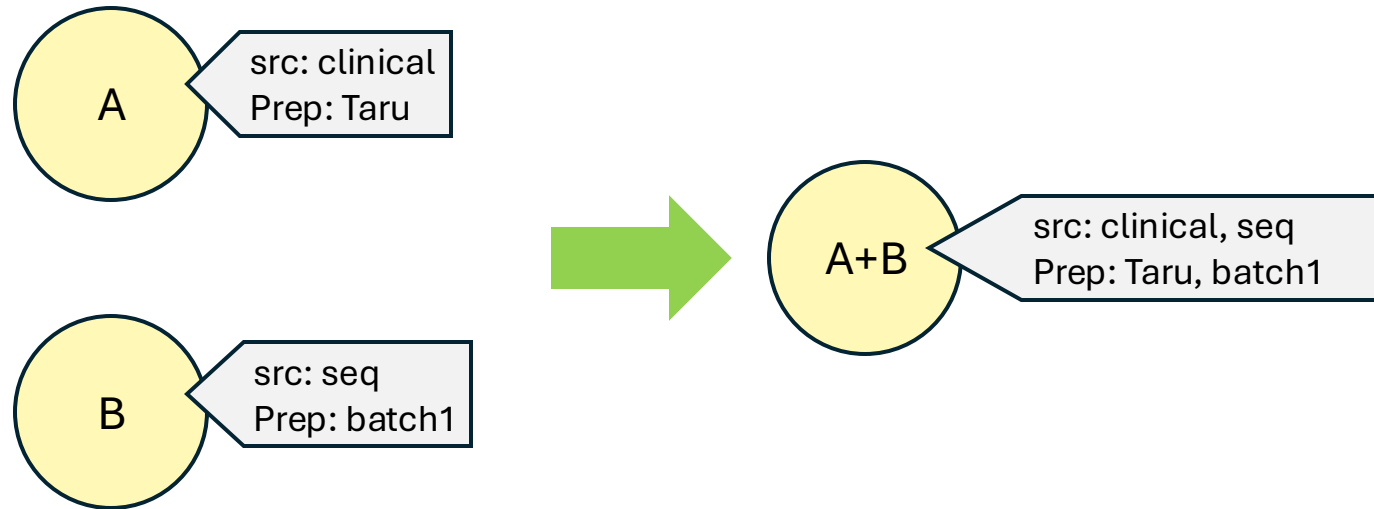
- What ?
 - Combine different sources of heterogeneous information,
- Why ?
 - Improve information quality
 - Improve SKG completeness
 - Provide a fuller and/or more precise understanding
 - Important for decision support systems
 - In contexts where information quality may be compromised
- How ?
 - Group together best quality information
 - Integrate information from diverse sources into a single SKG
 - Combine information so to get the best values for one problem

Information Fusion in Ontoweaver



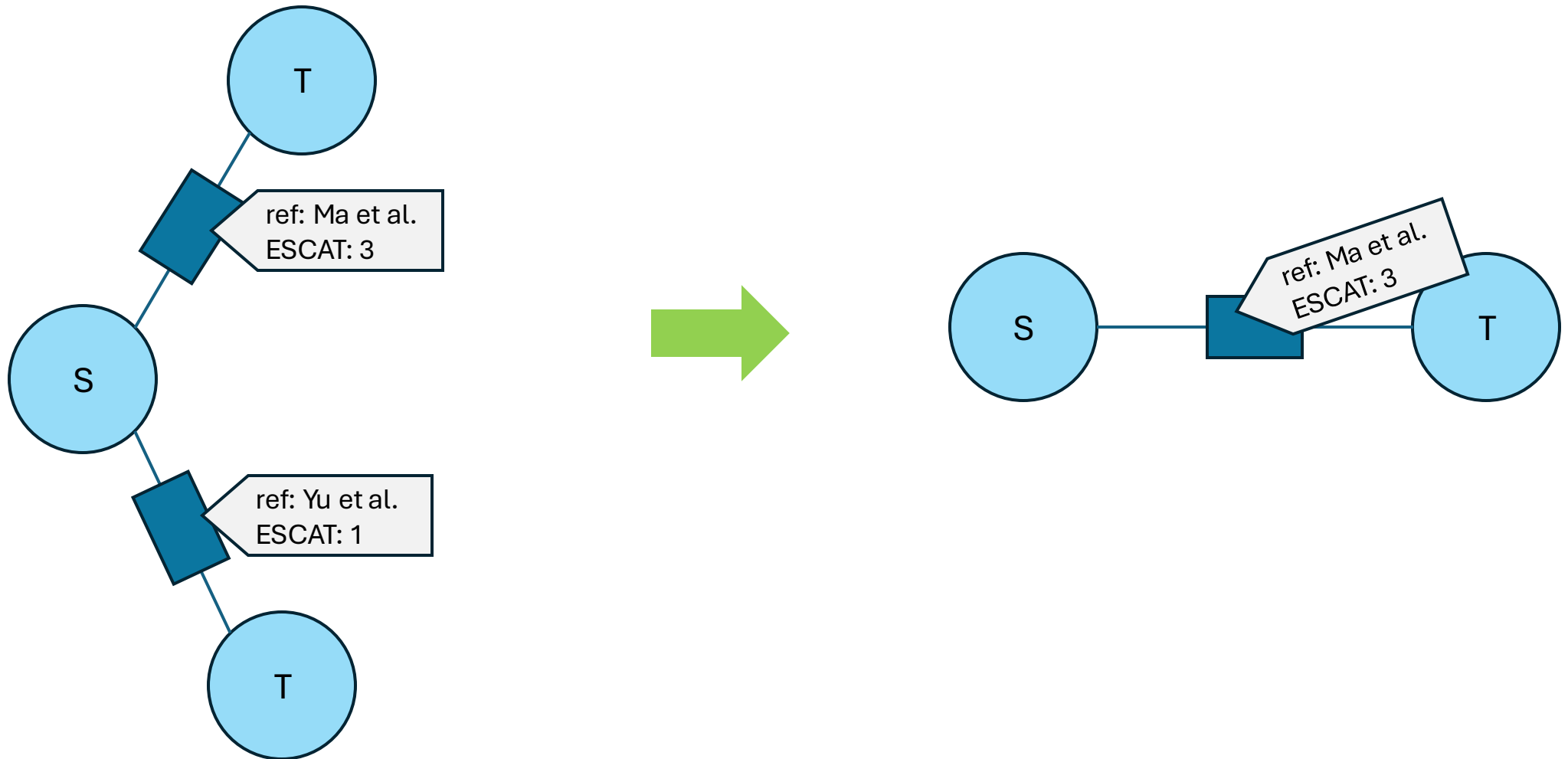
Examples of needs for information fusion

Different sources provide different observations



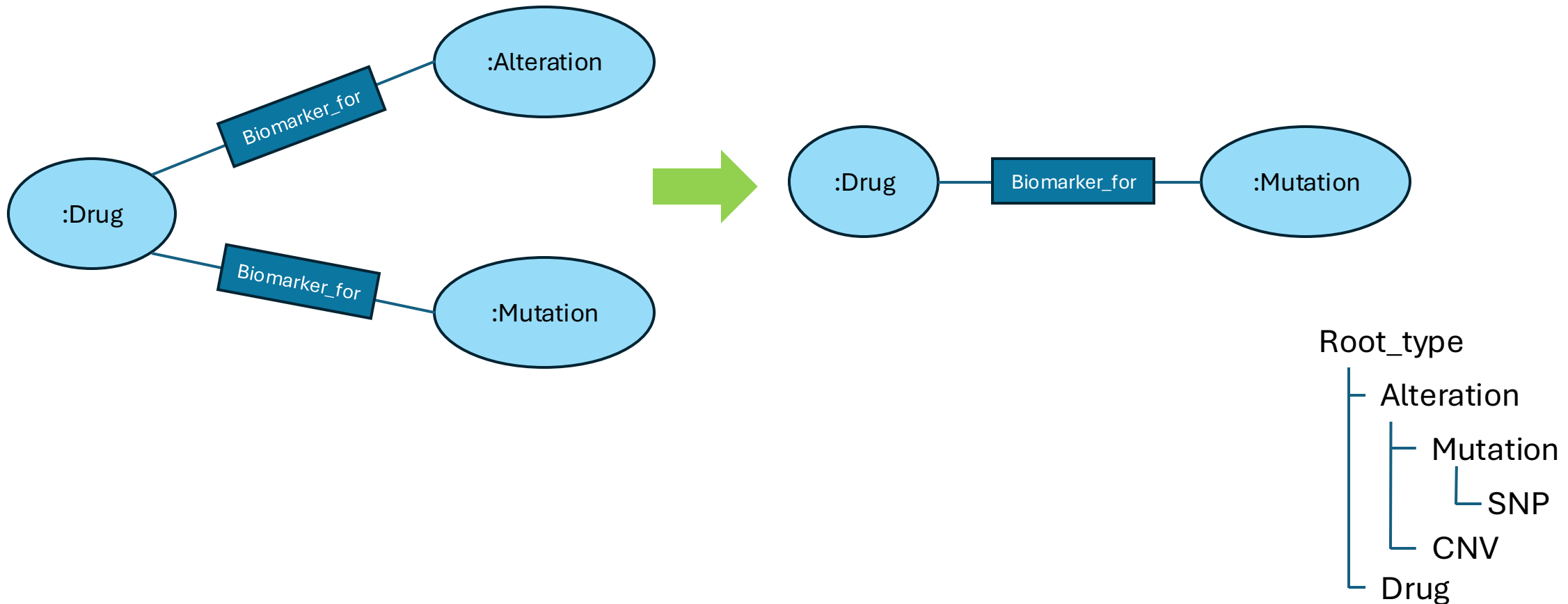
Examples of needs for information fusion

Different sources provide different level of confidence



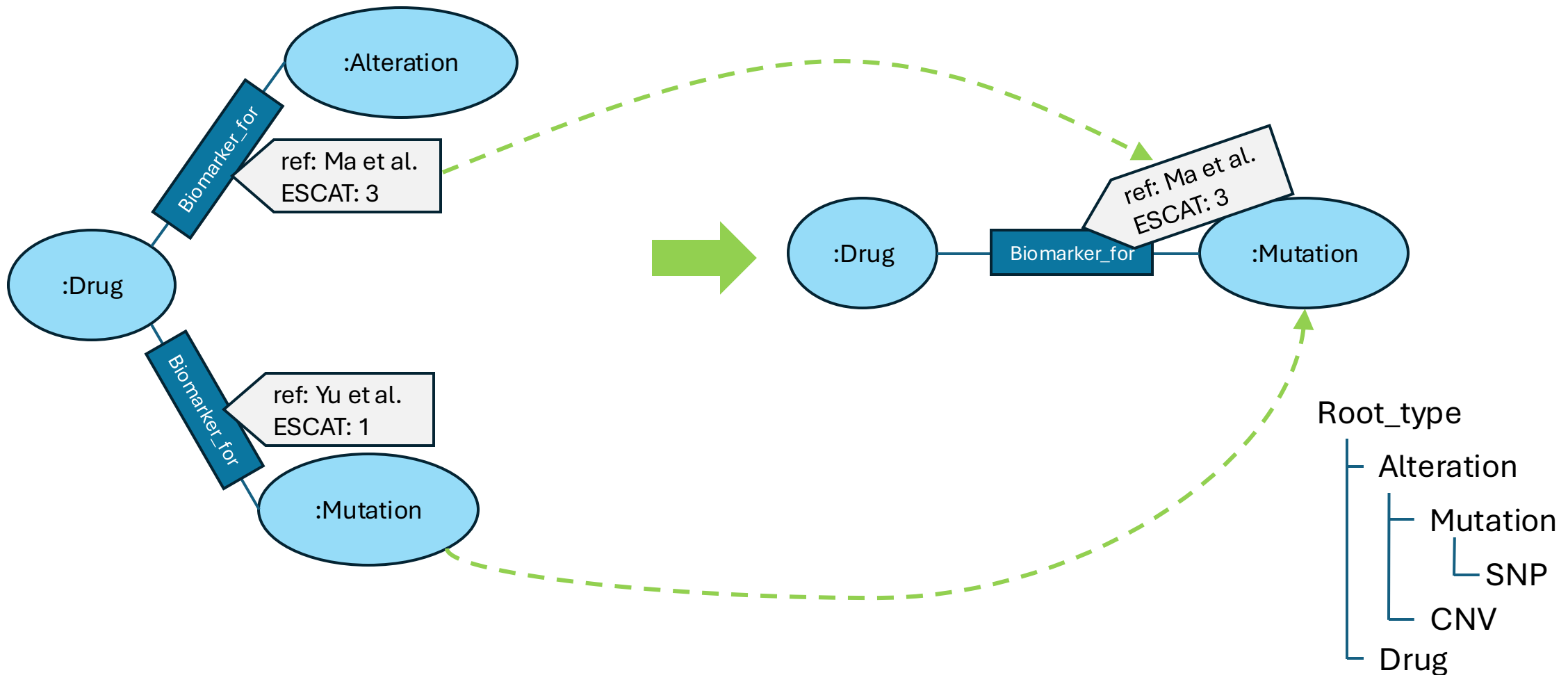
Examples of needs for information fusion

Different sources provide different level of precision



Examples of needs for information fusion

Different sources provide different level of precision



Deduplication (BioCypher) VS Fusion (OntoWeaver)?

- BC's deduplication:
 - Align with synonym_for
 - Keep data from the last seen ID
 - Supports large data graphs
- OW's fusion:
 - Align pieces of data, the way you want
 - Merge pieces of data, the way you want
 - Allows High-Level Information Fusion
 - Only for in-memory data (as of now)

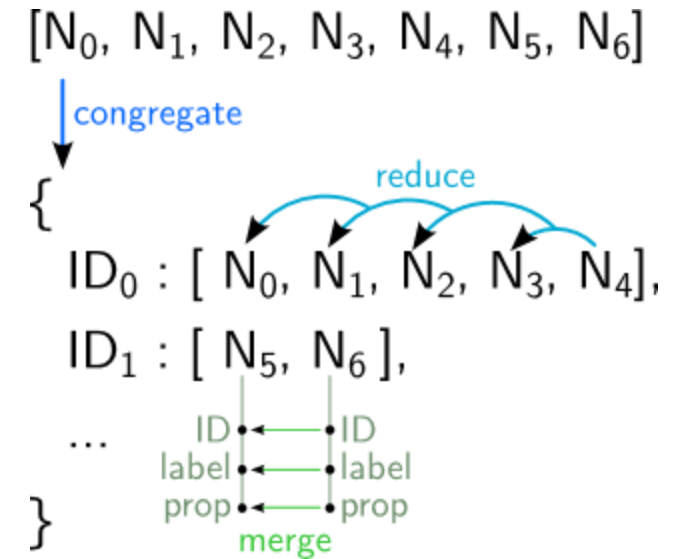
Fusion: a core feature of OntoWeaver

```
$ ontoweave  
| data_A1.csv:mapping_A.yaml  
| data_A2.csv:mapping_A.yaml  
| data_B/*.parquet:mapping_B.yaml  
| ontology.ttl:automap
```

Fusion (module) is **necessary** if one wants to apply several mappings **simultaneously**

Approach

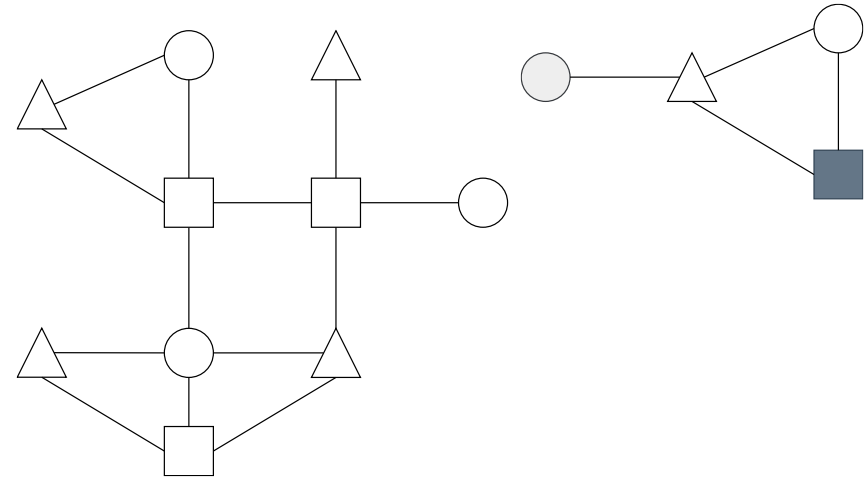
1. Identifier based alignment
 2. Iterative pairwise processing of nodes
 3. Pairwise property merge
- Different possible fusions
 - Default: add property values when several are available
 - Based on the source confidence
 - Based on the ontology hierarchy
 - ...



Semantic graph fusion

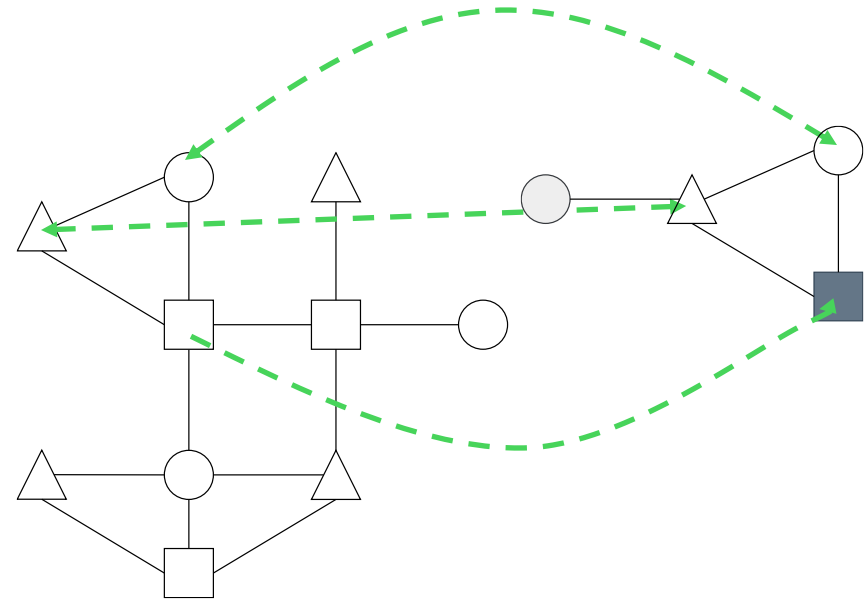
Semantic graphs fusion

- More generally, what is it ?
 - Alignment + Fusion



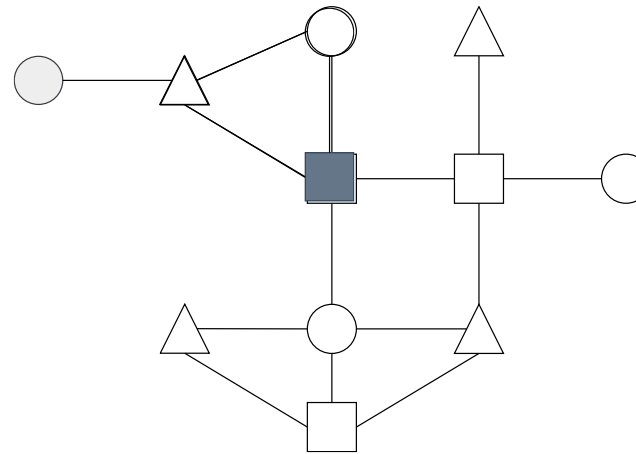
Semantic graphs fusion

- More generally, what is it ?
 - Alignment + Fusion
- Alignment :
 - Identity of nodes management
 - Subgraph isomorphism search



Semantic graphs fusion

- More generally, what is it ?
 - Alignment + Fusion
- Alignment :
 - Identity of nodes management
 - Subgraph isomorphism search
- Data fusion
 - At node/relation level
 - Property value enrichment



Nodes alignment

Alignment - Identity of nodes in KG

- Owl:sameAs / BioCypher's synonym_for
 - Many erroneous sameAs links
 - Pb of contextual identity

```
complex:
  synonym_for: macromolecular complex
  represented_as: node
  # ...
```

schema_config.yaml

```
PREFIX ex: <http://example.org>
PREFIX owl: <http://www.w3.org/2002/07/owl#>

# URI 1 from DBpedia
<http://dbpedia.org/resource/Sofia>
  ex:country "Bulgaria" .

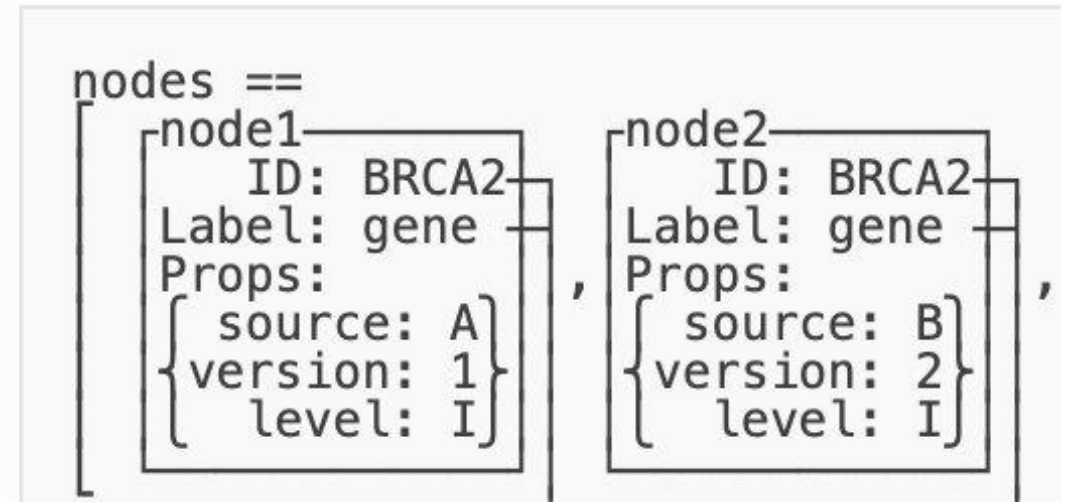
# URI 2 from GeoNames
<http://sws.geonames.org/727011/>
  ex:population 1200000 .

# Assertion stating both URIs are the same entity
<http://dbpedia.org/resource/Sofia>
  owl:sameAs <http://sws.geonames.org/727011/>
  .
```

ontology.ttl

Alignment - Identity of nodes in KG

- A Key uniquely identifies every instance of a class - By construction
 - Use of the “translate” transformer in the OW mapping



Alignment - Identity of nodes in KG

- Check if two nodes are “the same”
 - Based on the data without those nodes
 - Using rules, concatenation of properties...
 - e.g. homepage(x, y) and homepage(z,y) => sameAs(x,z)
- Default in ontoweave: serialization of ID+label in a string, test equality

```
nodes ==
[
  node1
  ID: BRCA2
  Label: gene
  Props:
  { source: A
    version: 1
    level: I
  },
  node2
  ID: BRCA2
  Label: gene
  Props:
  { source: B
    version: 2
    level: I
  },
  node3
  ID: BRCA2
  Label: prot
  Props:
  { source: B
    version: 2
    level: II
  },
  node4
  ID: BRCA2
  Label: gene
  Props:
  { source: B
    version: 3
    level: I
  }
]

>>> on_IDLabel = ontoweaver.serialize.IDLabel()
>>> for n in nodes:
>>>   on_IDLabel(n)

keys = ["BRCA2gene" , "BRCA2gene" , "BRCA2prot" , "BRCA2gene"]

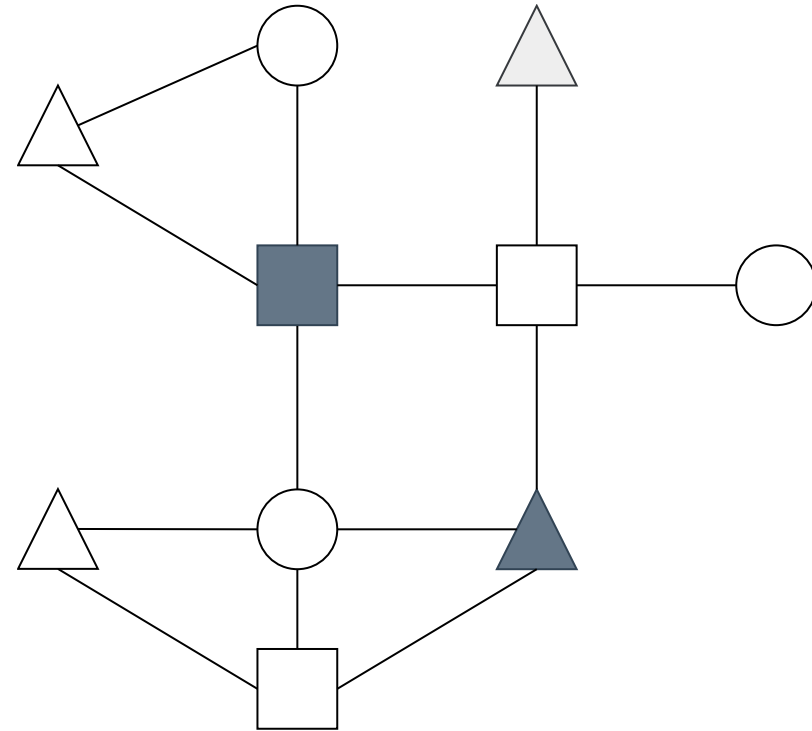
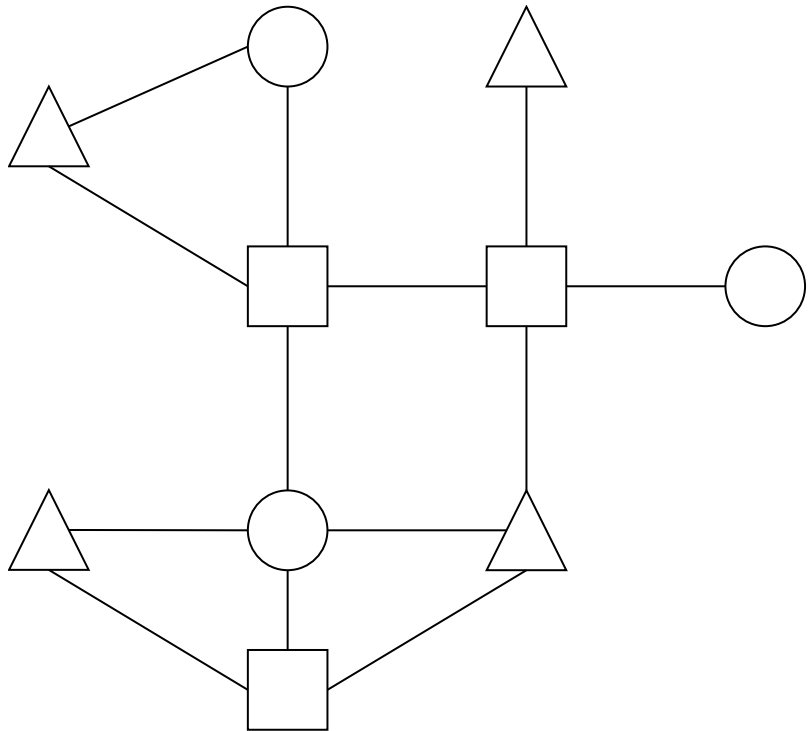
>>> congregate = ontoweaver.congregate.Nodes(on_IDLabel)
>>> congregate(nodes)

congregate.duplicates() ==
{
  "BRCA2gene": [
    node1
    ID: BRCA2
    Label: gene
    Props:
    { source: A
      version: 1
      level: I
    },
    node2
    ID: BRCA2
    Label: gene
    Props:
    { source: B
      version: 2
      level: I
    },
    node4
    ID: BRCA2
    Label: gene
    Props:
    { source: B
      version: 3
      level: I
    }
  ],
  "BRCA2prot": [
    node3
    ID: BRCA2
    Label: prot
    Props:
    { source: B
      version: 2
      level: II
    }
  ]
}
```

Alignment – Graph pairing

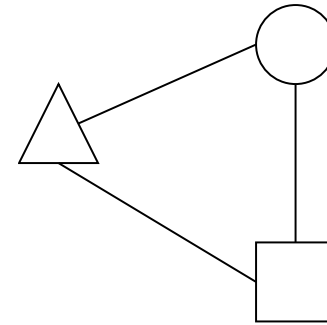
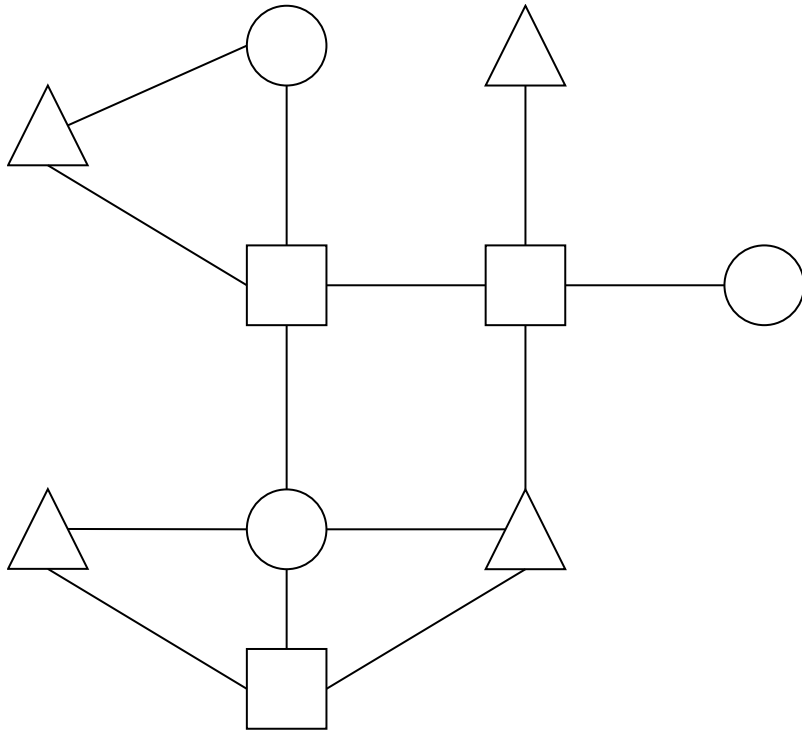
- Context:
 - When looking for the most generic approach
 - Uniqueness of ids cannot be ensured
- Difficulty in finding the right alignment
 - All nodes must be compared pairwise, given their properties
 - The relations between nodes must be aligned
- Subgraph isomorphism search
 - NP-complete problem

Subgraph isomorphism search



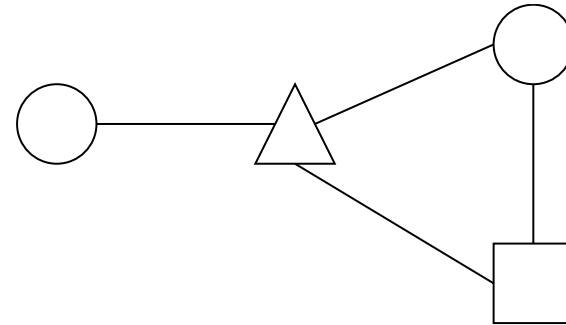
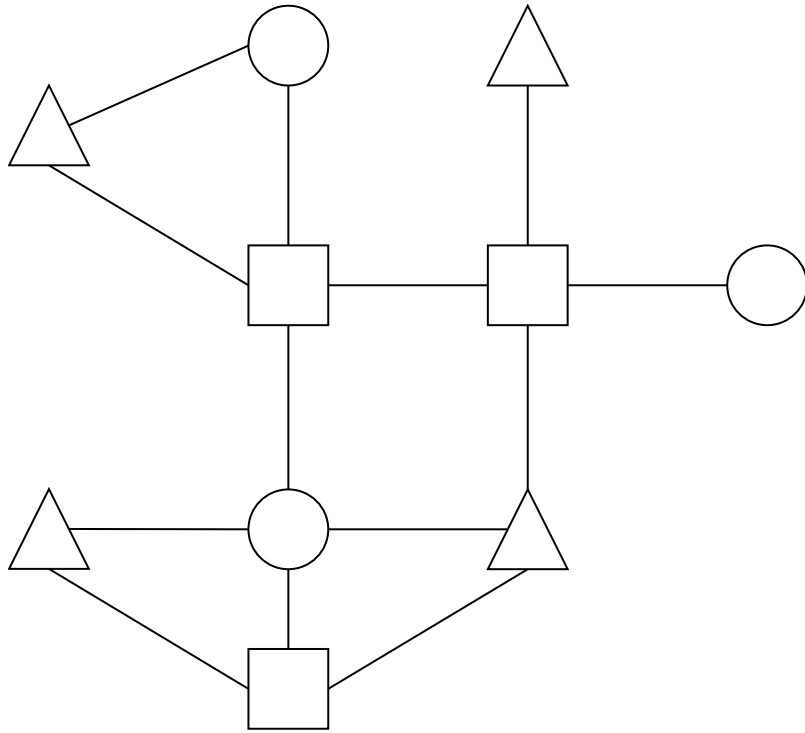
2 isomorphic graphs with different values in the nodes

Subgraph isomorphism search



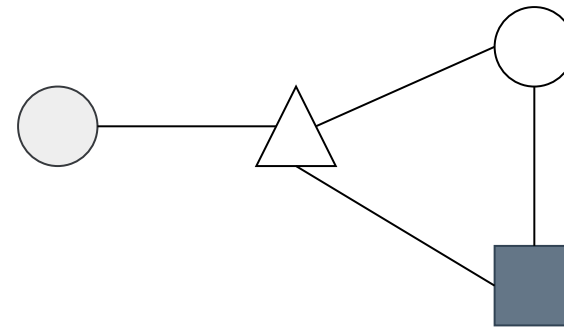
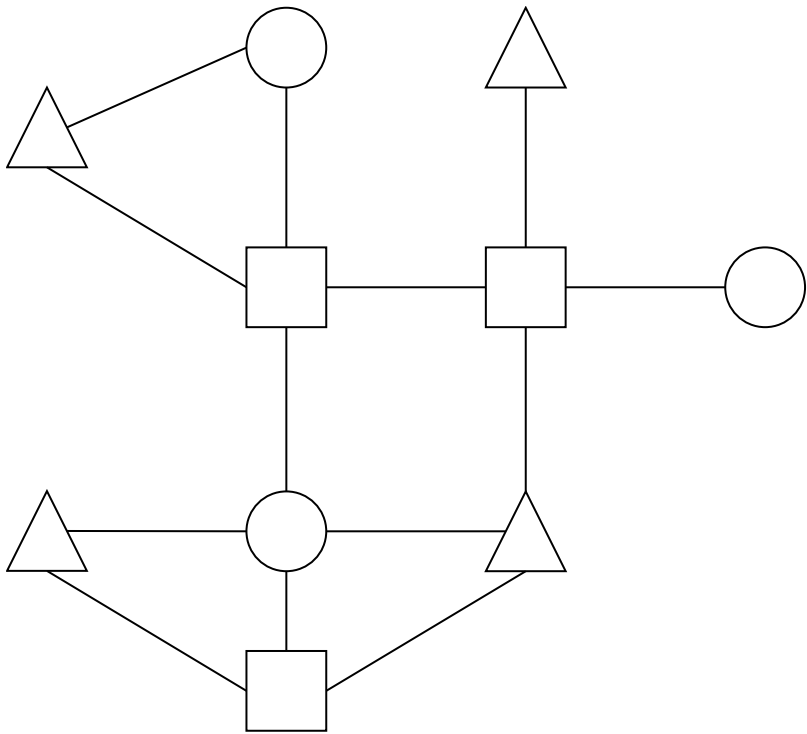
A "query" included in the SKG
NP-complete problem

Subgraph isomorphism search



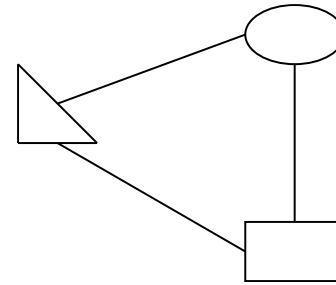
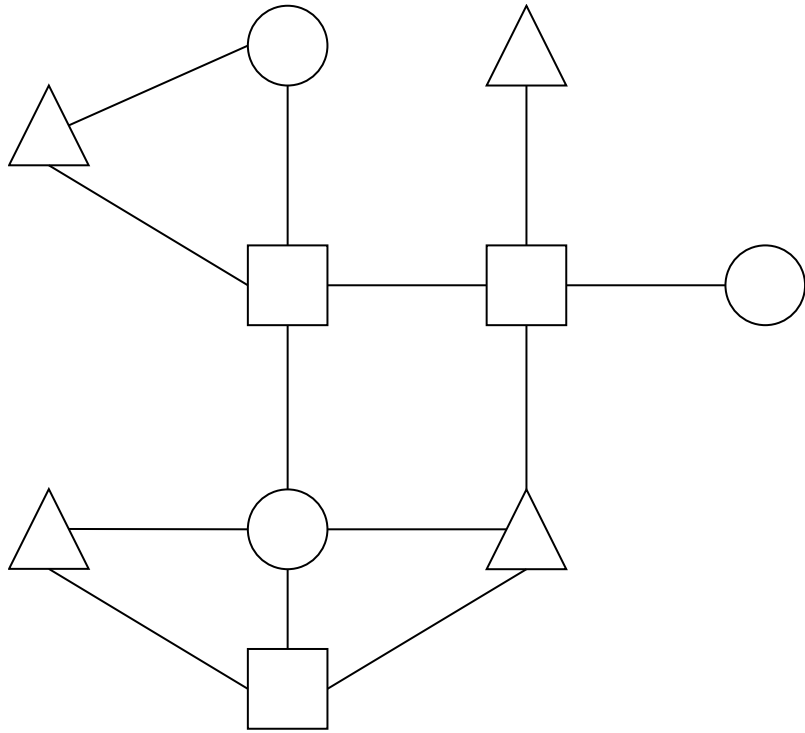
2 non isomorphic graphs with identical subgraphs

Subgraph isomorphism search

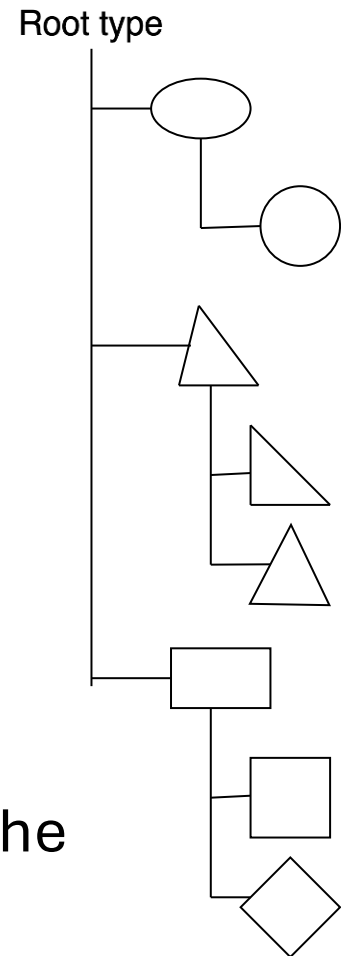


2 non isomorphic graphs with compatible sub-graphs (i.e. compatible node values)

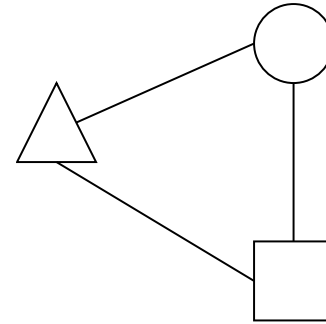
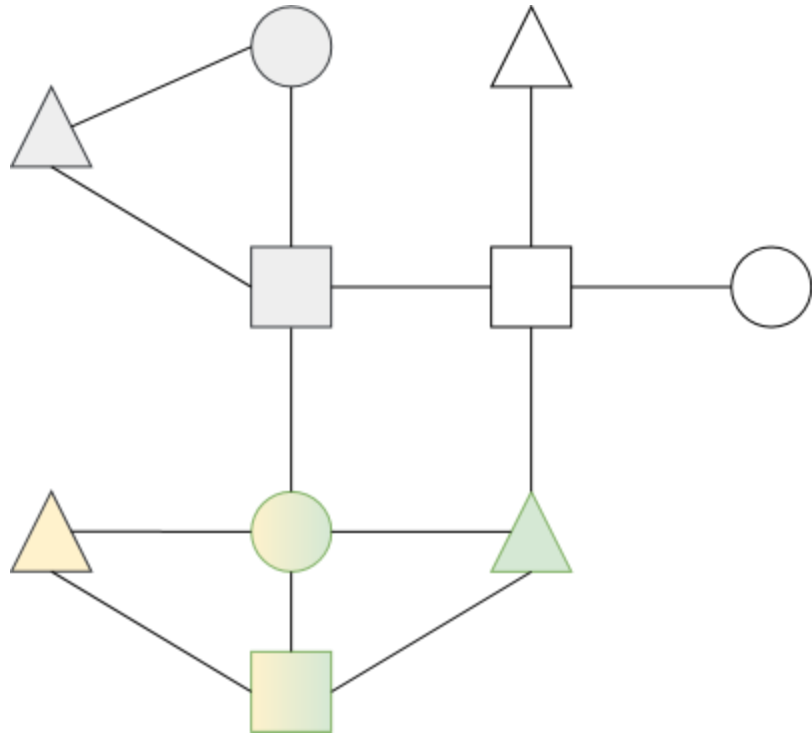
Subgraph isomorphism search



Even more difficult when taking into account the type hierarchy in the compatibility test



Subgraph isomorphism search



Even more difficult if we want all the possible alignments, in order to let the expert choose the best one

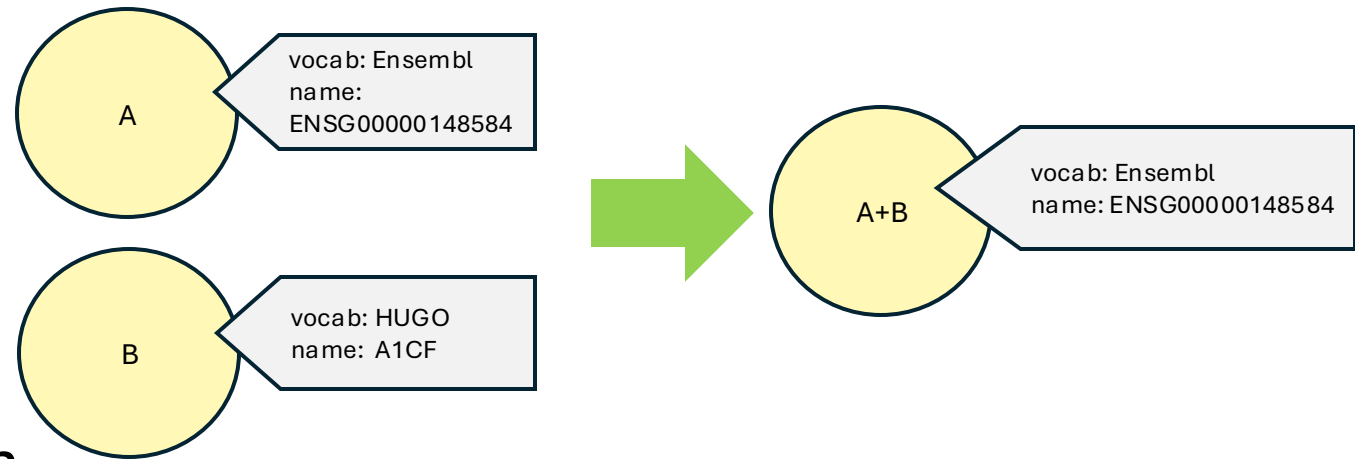
Subgraph isomorphism search

- Different implementations exist that can be implemented/used in the alignment/fusion module
 - Constraint Programming
 - Glasgow, VF2/VF3, LAD
 - Large-scale graph querying
 - GPU optimized solutions
 - ...
- Implementations in NetworkX
 - ISMAGS & VF2 algorithms

Nodes & Relations Merging

Nodes & relations fusion

- Example of conflicts
 - Data freshness
 - Different vocabularies
- Different strategies for fusion
 - Several OW modules

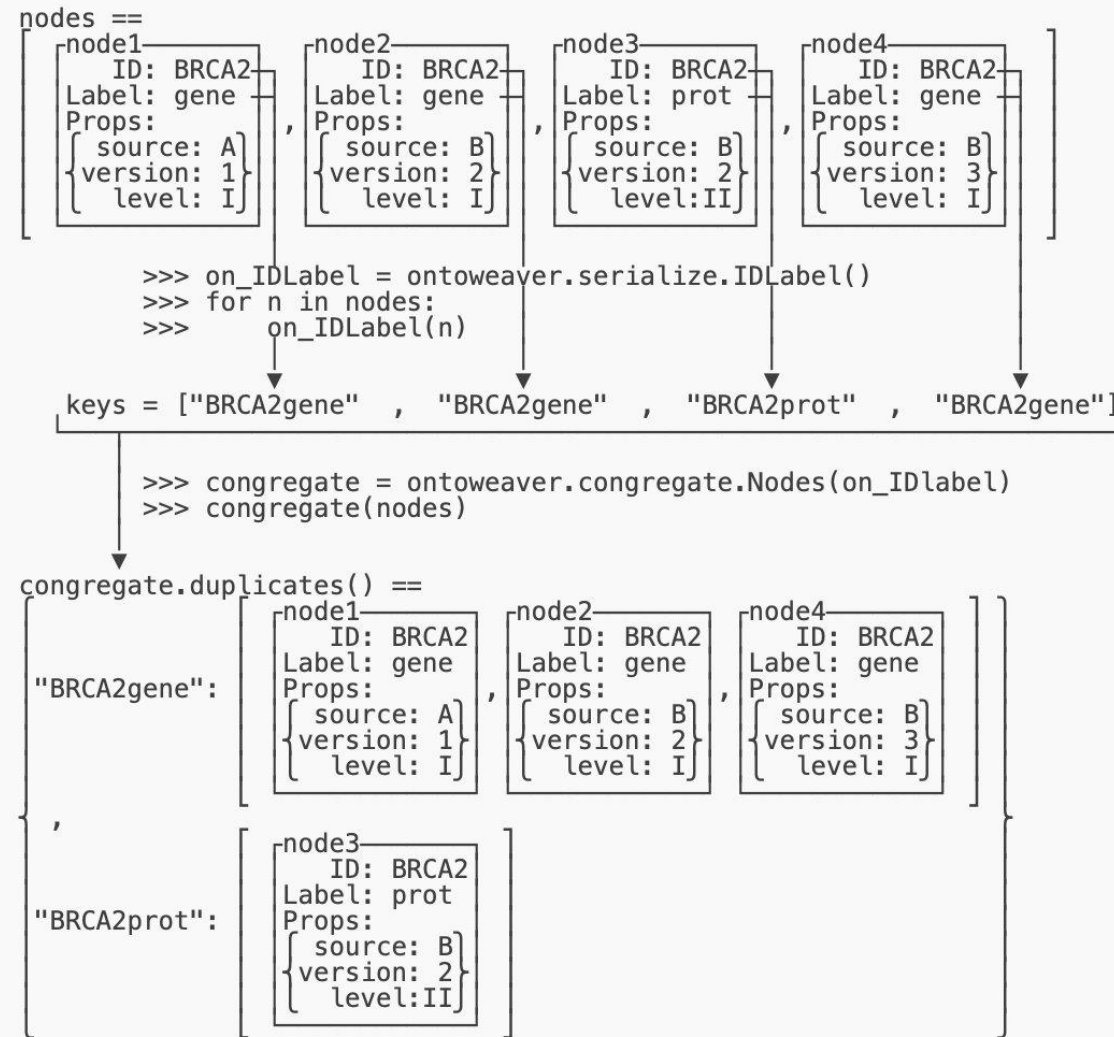


Need to keep track of the fusion “choices” and provide human readable explanations for experts

Nodes & relations fusion

- Data quality independent strategies
 - Most frequent value,
 - Aggregation functions
 - average, max, min, ...
- Data quality driven strategies
 - Data/Source confidence
 - Source reliability
 - Vote
 - Expert knowledge
 - Multi-valued properties

Example of Fusion in OntoWeaver

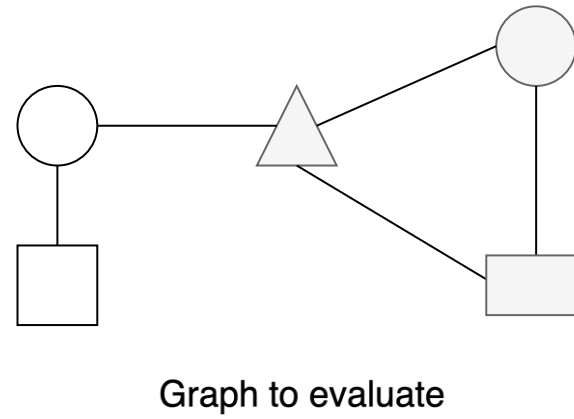
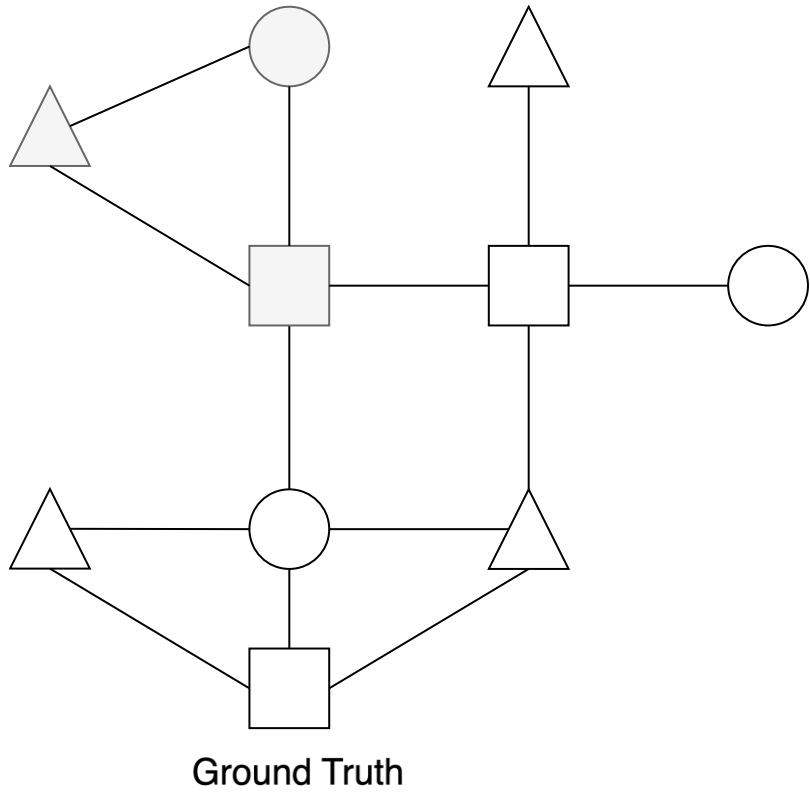


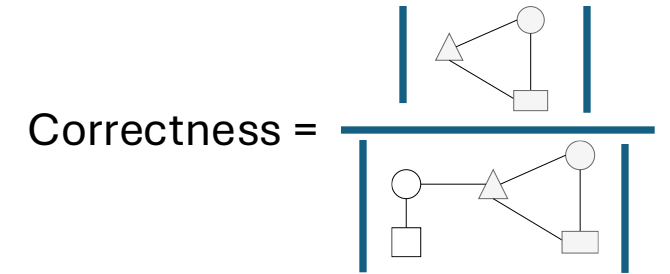
Quality metrics

Metrics to measure the quality of a fused graph

- General metrics on graphs that can give insights
 - Structure based: Size and distribution of data
- Evaluation of answers of a query
 - High-Level Information Fusion metrics
 - 3 complementary metrics
 - Correctness
 - Completeness
 - Precision
 - <https://github.com/clairelaudy/hlif-metrics>

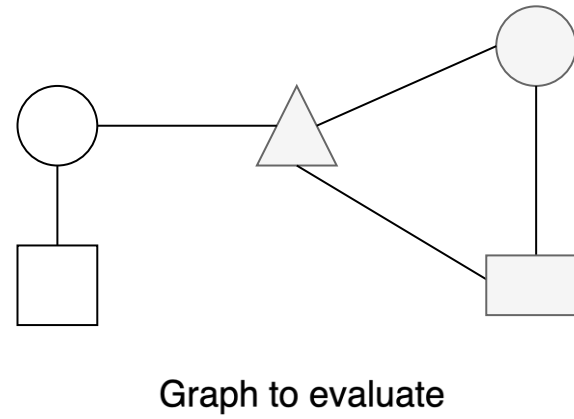
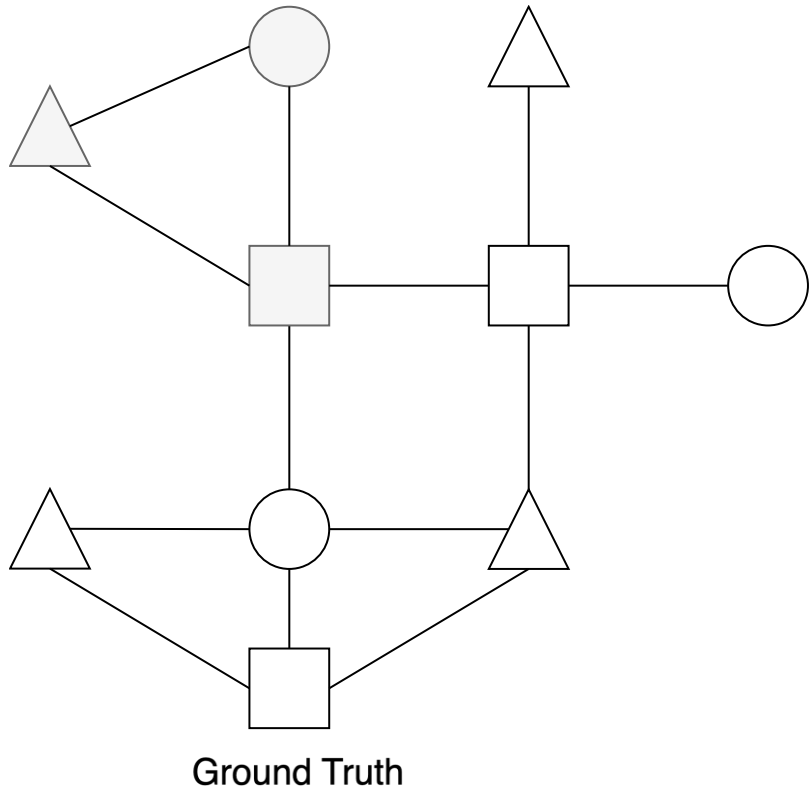
High Level Information Fusion metrics



$$\text{Correctness} = \frac{\text{Number of correct edges}}{\text{Total number of edges in the graph to evaluate}}$$


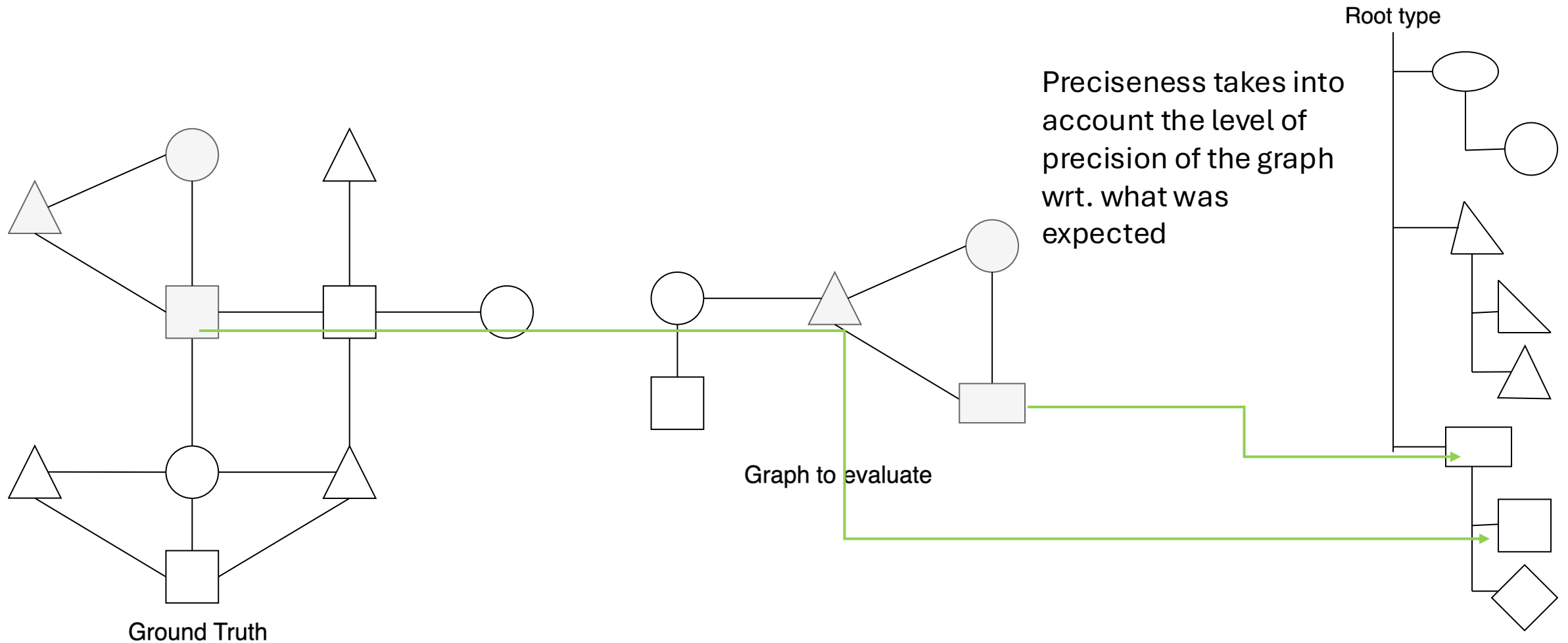
The diagram illustrates the Correctness metric. It shows a fraction where the numerator is the number of correct edges (3) and the denominator is the total number of edges in the graph to evaluate (6). The correct edges are highlighted in blue. The graph to evaluate is shown in the denominator, and the correct edges are highlighted in blue. The correct edges are: the edge between the top-left circle and the middle-left square, the edge between the middle-left square and the middle-right square, and the edge between the middle-right square and the rightmost circle.

High Level Information Fusion metrics



$$\text{Completeness} = \frac{\left| \begin{array}{c} \text{Top-left triangle} \\ \text{Top-left circle} \\ \text{Bottom-left square} \end{array} \right|}{\left| \begin{array}{c} \text{Top-left triangle} \\ \text{Top-left circle} \\ \text{Middle-left square} \\ \text{Middle-right square} \\ \text{Top-right triangle} \\ \text{Rightmost circle} \\ \text{Bottom-middle circle} \\ \text{Bottom-left triangle} \\ \text{Bottom-right triangle} \\ \text{Bottom-left square} \\ \text{Bottom-right square} \end{array} \right|}$$

High Level Information Fusion metrics

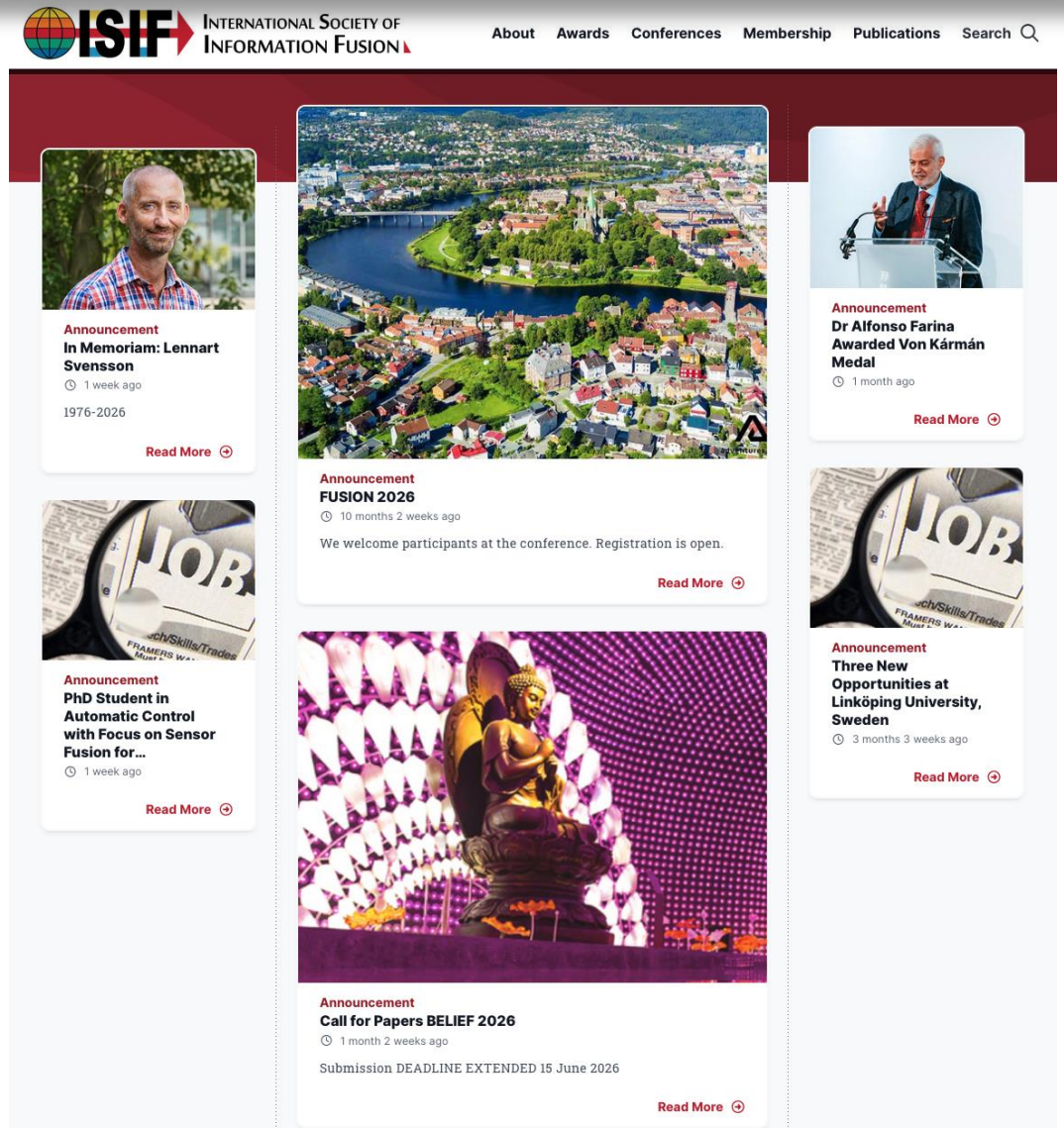


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ISIF – International Society of Information Fusion

The ISIF is the premier global information resource for multidisciplinary approaches for **theoretical and applied Information Fusion technologies.**

isif.org



The screenshot displays the ISIF website homepage. At the top, the ISIF logo is followed by the text "INTERNATIONAL SOCIETY OF INFORMATION FUSION". To the right of the logo is a navigation menu with links: "About", "Awards", "Conferences", "Membership", "Publications", and "Search". Below the navigation bar, the main content area features several announcement cards. Each card includes a header image, a title, a date, and a "Read More" link. The announcements include:

- Announcement In Memoriam: Lennart Svensson** (1 week ago, 1976-2026)
- Announcement FUSION 2026** (10 months 2 weeks ago, registration is open)
- Announcement PhD Student in Automatic Control with Focus on Sensor Fusion for...** (1 week ago)
- Announcement Dr Alfonso Farina Awarded Von Kármán Medal** (1 month ago)
- Announcement Three New Opportunities at Linköping University, Sweden** (3 months 3 weeks ago)
- Announcement Call for Papers BELIEF 2026** (1 month 2 weeks ago, submission deadline extended to 15 June 2026)

ISIF conferences & publications

- International Conference on Information Fusion
- Sponsored events
 - Conf. on Cognitive and Computational Aspect of Situation Management (COGSIMA)
 - *Biocypher User Workshop 2026*
 - Int. Conf. on Belief Functions
 - Latin American Workshop on Information Fusion (LAFUSION)
- Journal of Advances in Information Fusion
- ISIF supported Working groups
 - ETURWG – **Evaluation of Techniques for Uncertainty Representation** Working Group
 - <https://eturwg.c4i.gmu.edu/>
 - URREF ontology: <https://github.com/adelfphi23/urref>